P1 - Exam

1st Midterm Exam

Time: 70 minutes
Total: 100 points

GENERAL INSTRUCTIONS

Complete both exercises in the provided Excel workbook, using the existing worksheet.
For each exercise, create:

1. A VBA macro that collects the required information, performs the calculations, determines the results, writes the worksheet output, and displays a final message.
2. A separate recorded macro that formats the corresponding report.

You must create four macros in total.
Only concepts covered in class may be used.
Do not use:

- Loops
- Arrays
- Custom functions
- Additional worksheets
- UserForms
- Concepts not covered in class

Assume all numerical inputs are valid. All monetary inputs are nonnegative, and current revenue is greater than zero. Input validation is not required.
Perform the calculations in VBA and write the results to the specified cells.
Choose the appropriate decision structures yourself.
Save your completed workbook as an Excel Macro-Enabled Workbook (.xlsm) and also upload the Module file(s) (.bas).

EXERCISE 1: EMPLOYEE PAYROLL REPORT

Points: 50
A company needs to calculate an employee's monthly and annual compensation.
PART A: CALCULATION MACRO
Ask the user for:

- Employee name
- Monthly salary
- Monthly bonus
- Monthly deductions

Calculations:
Gross Monthly Compensation = Monthly Salary + Monthly Bonus
Net Monthly Compensation = Gross Monthly Compensation - Monthly Deductions

Annual Net Compensation = Net Monthly Compensation * 12
Compensation Classification:
Determine the classification using these rules:

- Annual net compensation of at least $600,000: "High Compensation"
- Annual net compensation of at least $300,000 but less than $600,000: "Medium Compensation"
- Annual net compensation below $300,000: "Standard Compensation"

Worksheet Output:
Write each label and its corresponding value in the following cells:
A1: Employee Payroll Report
A3: Employee Name
B3: Employee name
A4: Monthly Salary
B4: Monthly salary
A5: Monthly Bonus
B5: Monthly bonus
A6: Monthly Deductions
B6: Monthly deductions
A8: Gross Monthly Compensation
B8: Gross monthly compensation
A9: Net Monthly Compensation
B9: Net monthly compensation
A10: Annual Net Compensation
B10: Annual net compensation
A11: Classification
B11: Compensation classification
Final Message:
Display one message containing:

- Employee name
- Net monthly compensation
- Annual net compensation
- Compensation classification

PART B: RECORDED FORMATTING MACRO
Create a separate recorded macro that formats the report in A1:B11.
The formatted report must include:

- A bold title with a larger font size
- A background color for the title
- Bold labels
- Currency formatting with two decimal places for all monetary values
- Borders around the input and results sections
- A background color that distinguishes the results section
- Bold formatting for annual net compensation and classification
- Appropriate column widths so all content is visible

The formatting macro must preserve all report labels and values.

EXERCISE 2: DEPARTMENT PERFORMANCE REPORT

Points: 50
A company needs to evaluate a department’s profitability and spending.

PART A: CALCULATION MACRO

Ask the user for:

- Department name
- Current revenue
- Operating expenses
- Budgeted expenses

Calculations:

Operating Profit = Current Revenue - Operating Expenses

Profit Margin = Operating Profit / Current Revenue

Budget Difference = Budgeted Expenses - Operating Expenses

For this exercise, profit margin is a decimal ratio. For example, 0.20 represents 20%. Store the decimal ratio in the worksheet so the formatting macro can display it as a percentage.

Performance Classification:

Apply the following rules in order:

1. If operating profit is negative, classify it as "Loss-Making".
2. Otherwise, if profit margin is at least 20% and budget difference is greater than or equal to zero, the classification is "Strong".
3. Otherwise, if profit margin is at least 5%, the classification is "Acceptable".
4. Otherwise, classify it as "Weak".

Budget Status:

Determine the budget status:

- When operating expenses exceed budgeted expenses: "Over Budget"
- Otherwise: "Within Budget"

Worksheet Output:

Write each label and its corresponding value in the following cells:

D1: Department Performance Report

D3: Department Name

E3: Department name

D4: Current Revenue

E4: Current revenue

D5: Operating Expenses

E5: Operating expenses

D6: Budgeted Expenses

E6: Budgeted expenses

D8: Operating Profit

E8: Operating profit

D9: Profit Margin

E9: Profit margin as a decimal ratio

D10: Budget Difference

E10: Budget difference

D11: Classification

E11: Performance classification

D12: Budget Status

E12: Budget status

Final Message:

Display one message containing:

- Department name
- Operating profit

- Performance classification
- Budget status

PART B: RECORDED FORMATTING MACRO

Create a separate recorded macro that formats the report in D1:E12.

The formatted report must include:

- A bold title with a larger font size
- A background color for the title
- Bold labels
- Currency formatting with two decimal places for all monetary values
- Percentage formatting with two decimal places for profit margin
- Borders around the input and results sections
- A background color that distinguishes the results section
- Bold formatting for classification and budget status
- Appropriate column widths so all content is visible

The formatting macro must preserve all report labels and values.

EVALUATION CRITERIA

Each exercise is worth 50 points:

- Correct data input and variables: 7 points
- Correct calculations: 10 points
- Correct decision logic, including boundary values: 10 points
- Complete worksheet output in the specified cells: 8 points
- Correct final message: 5 points
- Recorded formatting macro meeting the requirements: 10 points

Total per exercise: 50 points

Total exam score: 100 points